

ITK-SNAP 4.6

Where the release stands, and what's coming for cardiac

Version 4.6.0-alpha.1 · Lab update, August 2026

Where 4.6 stands

Three things, and then the detail

1

What's already in

Open images and whole workspaces straight from a URL. Two AI-assisted segmentation models you can use today.

2

What's coming — cardiac first

Propagation across phases, 4D cardiac I/O, one-button automatic segmentation, and mesh tooling.

3

No release date yet

The two most visible cardiac features are waiting on one piece of server work. Slide 7 says which.

Open your data straight from a URL

The largest block of work in 4.6 — done and merged

Your study data stays where it lives.

ITK-SNAP now opens images — and whole workspaces — directly from a URL. No download step, and no local copy quietly drifting out of sync with the server.

A workspace with several layers opens over a single authenticated connection, and the second time you open it, it comes from the local cache instantly.

scp:// **sftp://**

Your lab server, over SSH

http:// **https://**

Public datasets, cached

fw://

Flywheel, with API-key auto-load

itksnap-fw://

A link that opens in a running ITK-SNAP

Two AI models today — not interchangeable

Which one you pick matters for cardiac work

nnInteractive

RECOMMENDED FOR CARDIAC

Works in	3D — scribble one slice, get the whole structure
You give it	point, box, scribble (paintbrush), lasso (polygon)
Image type	grayscale
Trained on	medical images (DKFZ, nnU-Net lineage)

SAM2

2D, POINT PROMPTS

Works in	2D — one slice at a time
You give it	a point
Image type	grayscale or colour
Trained on	natural images and video

For cardiac chambers and myocardium, use nnInteractive. It is 3D-native and medically trained — one scribble on a short-axis slice propagates through the stack.

Also new in 4.6

1

Work across windows

A Window menu listing every running ITK-SNAP, and Send to Other Window between them.

2

Cancel a running task

An X in the progress overlay stops a long download or computation.

3

More formats

.vti and .vtp I/O, and .seq.nrrd sequence export — contributed from our fork.

4

Clearer failure on big images

An explicit out-of-memory dialog instead of a hard failure.

Stability since 4.4

24

bugfixes

9

memory leaks

2

crashes fixed
last week

Including a server connection race and two long-standing DICOM crash and hang bugs.

One note on how: a lot of this was found by pointing AI coding agents at the codebase and at our own crash logs — including a regression that had been dormant since March.

What's coming — cardiac first

In the order they matter to this room

1

Propagation across phases

Segment one cardiac phase, propagate to the rest.

Waiting on server work

2

4D cardiac I/O

Phase and time handling for cine CTA and echo, plus sequence export.

On a branch — not yet merged

3

cmesh integration

Mesh operations move to a proper shared library.

Waiting on a release tag

4

One-button automatic segmentation

Pick a model, press once, get labelled cardiac anatomy — no prompts.

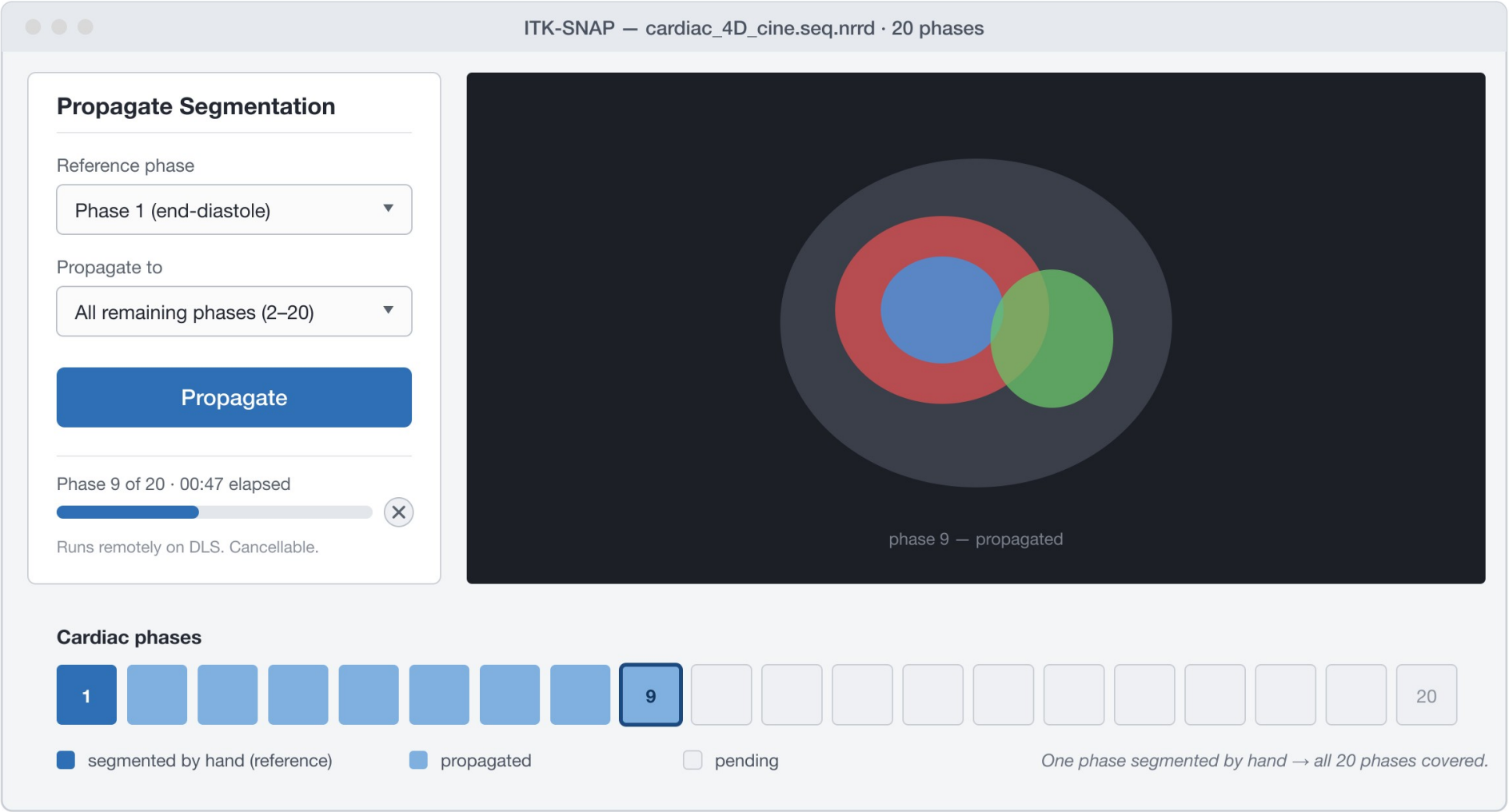
Waiting on server work

Also on the list: 3D/2D cursor sync under free rotation (a bug), developer documentation, and a rolling bugfix stream.

Worth saying once: propagation and automatic segmentation wait on the same piece of server work. It is already written — it needs promoting, not inventing.

What propagation will look like

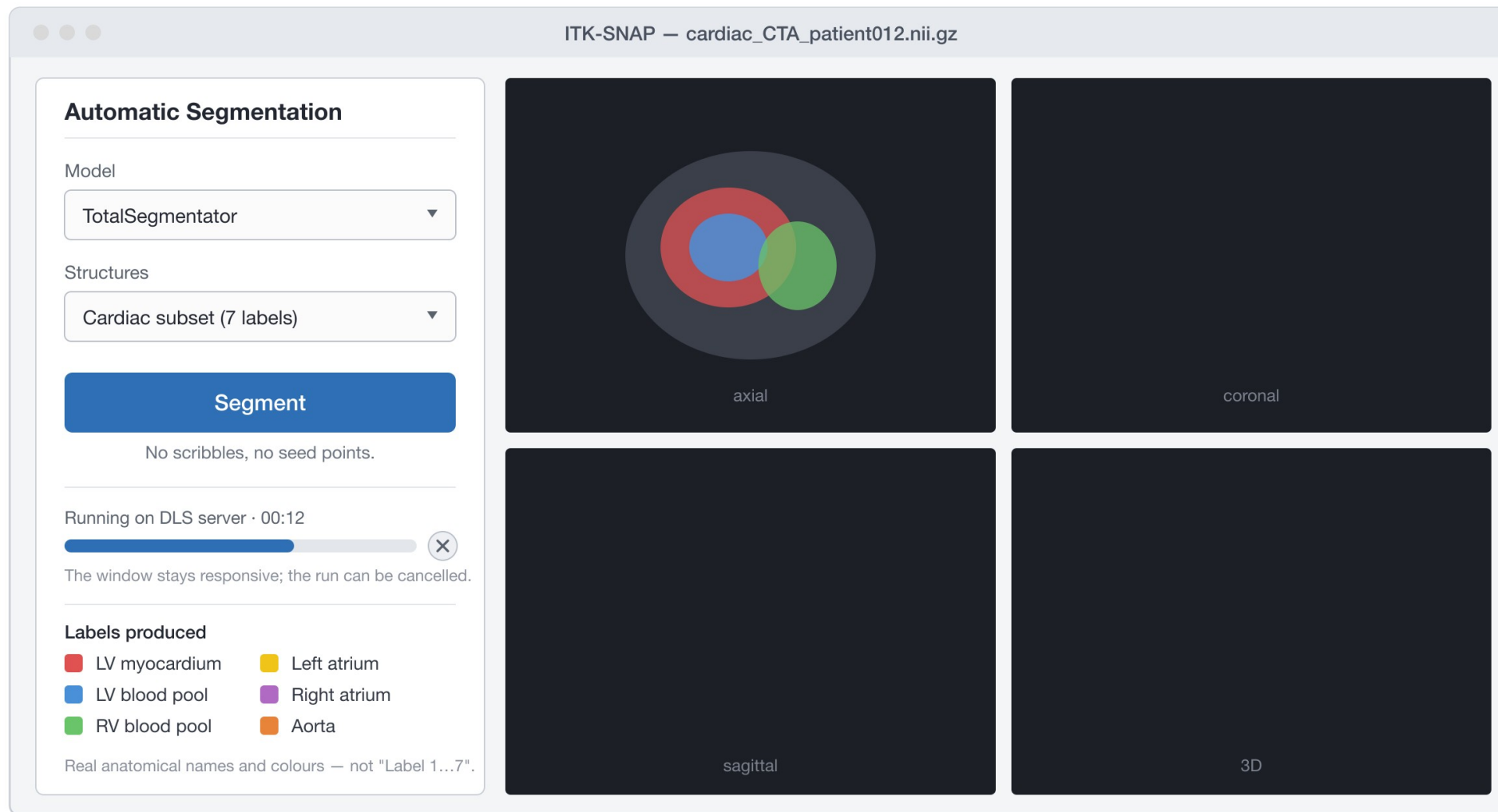
CONCEPT MOCK-UP



Segment one phase by hand. Press Propagate. The rest are filled in — on the server, with progress, cancellable.

What automatic segmentation will look like

CONCEPT MOCK-UP



Open the image, pick a model, press Segment — labelled cardiac anatomy with real names and colours.

Demo

Pre-recorded — because this is a network feature, and I am not gambling a live SSH handshake on the room's wifi.

What you'll see

About five minutes

- 1 Paste a Flywheel URL — the image opens, with no download step
- 2 Cancel a load mid-download from the progress overlay
- 3 Open a multi-layer workspace over one connection
- 4 **nnInteractive: scribble one short-axis slice, get the 3D chamber**
- 5 Send an image to a second ITK-SNAP window
- 6 Re-open the same remote image — instant, from cache

What's next

The server refactor — it unblocks propagation and automatic segmentation together.

What I need from you

- If you use Flywheel — try the fw:// path and tell me what breaks.
- Cardiac datasets I can test propagation against.

No release date yet. The next checkpoint is that server work landing.

One more thing...

ITK-SNAP, callable by an AI agent

A research prototype — not part of 4.6, and not on the 4.6 roadmap

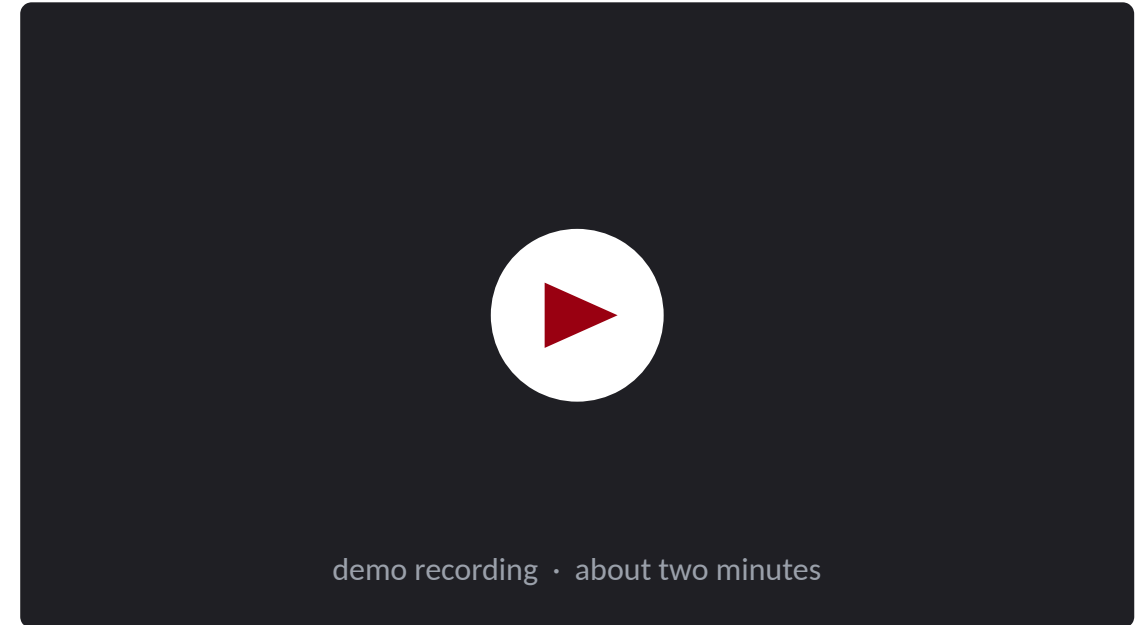
Model proposes, human disposes.

An AI agent can drive ITK-SNAP over the Model Context Protocol — open a study, run a model, apply a segmentation, hand the uncertain cases to a person.

And every edit comes back as a structured record of what changed and who changed it — the agent, or you.

Submitted to the SIIM-CAIMI26 AI Builder Showcase.

Hao · Pouch · Yushkevich. Notification late August; conference is 26–27 October, at Penn.



- 1 The agent calls ITK-SNAP as a tool 45s
- 2 **Model proposes, human disposes — the live handoff** 60s
- 3 The correction is a return value 30s